PYTHON • LECTURE 05

Loops in Python

while • for • range() • break • continue

Instructor

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Class Focus

Concepts • Code • Practice

Learning Objectives

KEY IDEA

Understand while and for loops; use range(); control loops with break and continue; solve basic loop problems.

What is a Loop?

KEY IDEA

A loop repeats a block of code. It reduces repetition and is useful when the same action must run multiple times.

Two Main Loops in Python

while LOOP

Repeats as long as a condition remains True.
Useful when the number of repetitions is not fixed.

for LOOP

Iterates through an iterable such as a list, tuple, string or range.

while Loop — Basic Syntax

```
while condition:
    statement(s)
```

while Loop — Print 1 to 5

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Understanding the Counter

1
Initialize

2
Check

3
Execute

4
Update

5
Repeat

Print 5 to 1

```
i = 5
while i >= 1:
    print(i)
    i -= 1
```

Infinite Loop

KEY IDEA

If the loop condition never becomes False, the loop continues indefinitely. Always ensure the condition can change.

Practice — Print Numbers

```
i = 1
while i <= 100:
    print(i)
    i += 1
```

Practice — Multiplication Table

```
n = int(input('Enter number: '))
i = 1
while i <= 10:
    print(n * i)
    i += 1
```

Iterating Through a List

```
nums = [1, 4, 9, 16, 25]
i = 0
while i < len(nums):
    print(nums[i])
    i += 1
```

Search for a Number in a Tuple

```
nums = (1, 4, 9, 16, 25, 36)
x = 16
i = 0
while i < len(nums):
    if nums[i] == x:
        print('Found at index', i)
    i += 1
```

break Statement

```
i = 1
while i <= 10:
    if i == 5:
        break
    print(i)
    i += 1
```

continue Statement

```
i = 0
while i < 5:
    i += 1
    if i == 3:
        continue
    print(i)
```

for Loop

KEY IDEA

A for loop is used to iterate through strings, lists, tuples and other iterable objects.

for Loop with a List

```
nums = [1, 2, 3, 4, 5]
for val in nums:
    print(val)
```

for Loop with a String

```
name = 'Python'
for ch in name:
    print(ch)
```

range() Function

range(stop)

Starts at 0 and stops before stop.
Example: range(5) → 0,1,2,3,4

range(start, stop)

Starts at start and stops before stop.

range(start, stop, step)

Controls the increment/decrement using step.

range(stop)

```
for i in range(5):
    print(i)
```

range(start, stop)

```
for i in range(1, 6):
    print(i)
```

range(start, stop, step)

```
for i in range(2, 11, 2):
    print(i)
```

Practice — Even Numbers

```
for i in range(2, 101, 2):
    print(i)
```

Practice — Multiplication Table with for

```
n = int(input('Enter number: '))
for i in range(1, 11):
    print(n, 'x', i, '=', n*i)
```

pass Statement

```
for i in range(5):
    pass

print('Program continues')
```

Practice — Sum of First n Numbers

```
n = int(input('Enter n: '))
sum = 0
for i in range(1, n+1):
    sum += i
print('Sum =', sum)
```

Practice — Factorial

```
n = int(input('Enter n: '))
fact = 1
for i in range(1, n+1):
    fact *= i
print('Factorial =', fact)
```

while vs for

while

Condition-controlled
Manual update often required
Useful for unknown repetition count

for

Iterable-controlled
Concise syntax
Useful for sequences and known ranges

Class Challenge

Task 1

Print 1–10

Task 2

Odd numbers 1–20

Task 3

Table of 7

Task 4

Sum 1–50

Task 5

Search an item in a list

Lecture Recap

while

Repeat while a condition is True.

for + range()

Iterate through sequences and generated numbers.

Loop Control

break exits; continue skips; pass is a placeholder.